

GENERICS

// class
class Box<T> { }
// variable
Box<String> b = new Box<String>();
// method
public <T> void doStuff(T value) { }

JAVA BEING STUPID

T obj = new T(); // error
obj instanceof T; // error
T heh = (T) "asdf"; // no typechecking
var s = Stack<int>(); // no primitive types
private static T x; // static no worky
Node<Number> s = new Node<Integer>; // error
class IThrowAnErrorBecauseWhyNot {
 void m(List<String> list) { }
 void m(List<Integer> list) { }
} // error because of type erasure & identifiability
Integer[] c = set.toArray(new Integer[0]);

ITERATOR

for (String s : stringList) { // ==
 for (Iterator<String> i = stringList.iterator();
 i.hasNext();) {
 String s = i.next();
 }
}

ITERABLE

interface Iterable<T> {
 Iterator<T> iterator();
}
interface Iterator<E> {
 boolean hasNext();
 E next();
}

COMPARABLE

TODO:
Type signature?

static <T extends Comparable> T doStuff(T a, T b) {
 // compareTo is possible with all types
 return a.compareTo("b") > 0 ? a : b;
}
static <T extends Comparable<T>> T doStuff(T a, T b)
{
 // compareTo is possible only with T
 return a.compareTo(b) > 0 ? a : b;
}

EXAMPLE

class Graphic {
 public void draw() { }
}
class Rectangle extends Graphic { }
class Circle extends Graphic { }
class GraphicStack<T extends Graphic>
 extends Stack<T> {
 public void drawAll() {
 for (T item : this) item.draw();
 }
}

BYTECODE

TODO:

TYPE ERASURE

- Introduced because of "bAcKwArDs CoMpAtAbIlItY"
- No type information at runtime (Non-Reifiable Type)
class MyStack<T> {
 Entry<T> top;
 int push(T value) { }
}
// becomes
class MyStack {
 Entry top;
 int push(Object value) { }
}
/* with bounds */
class MyStack<T extends Comparable<T>> {
 Entry<T> top;
 int push(T value) { }
}
// becomes

class MyStack {
 Entry top;
 int push(Comparable value) { }
}
/* what happens at/before runtime */
MyStack<String> s = new MyStack<String>();
s.push("Eh");
String x = s.pop();
// becomes
MyStack s = new MyStack();
s.push("Eh");
String x = (String) s.pop();

WILDCARDS

void printGs(List<Graphic> gs) { }
List<Graphic> gs1 = new ArrayList<>();
printGs(gs1); // ok
List<Circle> gs2 = new ArrayList<>();
printGs(gs2); // error
/* solution */
void printGs(List<? extends Graphic> gs) { }

GENERIC VARIANCE

	Type	Compatible Type-Args	R	W
Invariance	C<T>	T	✓	✓
Covariance	C<? extends T>	T and sub-types	✓	×
Contravariance	C<? super T>	T and basis-types	×	✓
Bivariance	C<?>	All	×	×

INVARIANCE

static <T> void move(Stack<T>, from, Stack<T> to) {
 while(!from.isEmpty()) to.push(from.pop());
}
var rectS = new Stack<Rectangle>();
var graphS = new Stack<Graphic>();
graphS.push(new Rectangle()); // ok
move(rectS, graphS); // error
Stack<Object> objS = graphS; // error
/* anotherone */
String[] strA = new String[10];
Object[] objA = strA; // ok
objA[0] = Integer.valueOf(2); // runtime err

TODO:

// compiles?
List<Graphic> list = new ArrayList<Rectangle>();
// compiles?
Rectangle[] rectangleStack = new Rectangle[10];
Graphic[] graphicStack = new Graphic[10];
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
graphicStack = rectangleStack;
// compiles?
Stack<Rectangle> rectangleStack = new Stack<>();
Stack<Graphic> graphicStack = new Stack<>();
graphicStack = rectangleStack;
for (Rectangle r : rectangleStack) {
 graphicStack.push(r);
}

COVARIANCE

public class Stack<E> {
 public void pushAll(Iterable<? extends E> src) { }
}
Stack<? extends Graphic> graphS;
graphS = new Stack<Rectangle>(); // ok
graphS = new Stack<Circle>(); // ok
graphS = new Stack<Object>(); // error (duh)
/* read only */
Stack<? extends Graphic> stack = new
 Stack<Rectangle>();
stack.push(new Graphic()); // error
stack.push(new Rectangle()); // error
stack.push(new Triangle()); // error

CONTRAVARIANCE

static <T> void addToC(List<? super T> list, T e) {
 list.add(e);
}
addToC(new ArrayList<Number>(), 3); // ok
addToC(new ArrayList<Object>(), 3); // ok

/* shenanigans */
Stack<? super Graphic> s = new Stack<Object>();
s.add(new Graphic()); // ok
s.add(new Rectangle()); // ok
Graphic g = s.pop(); // error
Object g = s.pop(); // ok

BIVARIANCE

static void printList(List<?> list) {
 for (Object elem : list)
 System.out.print(elem + " ");
}
/* more shenanigans */
static void appendNewObject(List<?> list) {
 list.add(new Object()); // error
}

PRODUCER / CONSUMER

TODO:

MISC

<T extends Comparable & Collection> // multiple type
 bounds
GENERIC STREAM TOMFOOLERY

List<? extends Media> medialis;
public <S extends T> List<S>
 filterBySubtype(Class<S> subtype) {
 return medialis.stream()
 .filter(subtype::isInstance)
 .map(subtype::cast)
 .collect(Collectors.toList());
 }
}

ANNOTATIONS

- Metadata
- Not actually part of the code

TODO:
slides 10

- @Override
- @Test
- @Json...

DEFINING

@Target(ElementType.TYPE)
@Retention(RetentionPolicy.RUNTIME)
public @interface Author {
 String name();
 String date() default "N/A";
}

@Author(name = "UwU", date = "idon't validate input")
public class WeeDoo { }

REFLECTON

- Information of Class (private and public): Methods, Attrib-
 utes, Parent class, Implemented interfaces
- Calling methods possible
- Usages: Debugger, Serialization, Frameworks, Remote
 Procedure Call, ORM
// all of these `throws SecurityException`
public Field[] getDeclaredFields()
public Method[] getDeclaredMethods()
public Constructor<?>[] getDeclaredConstructors()
System.out.println(dump(new Student("a", "b"), 0));
// {
// firstName = a
// lastName = b
// }
Class c = "s".getClass(); // java.lang.String

TODO:
Diagram (slides 26)

METHOD

@Target(ElementType.METHOD)

public String getName()
public boolean isAnnotationPresent(
 Class<? extends Annotation> annotationClass)
public Object invoke(Object obj, Object... args)

throws IllegalAccessException,
 IllegalArgumentException,
 InvocationTargetException
public class Profiler {
 public static void main(String[] args) {
 var testFunctions = new ProfileFunctions();
 int[] array = {5, 2, 8, 1, 93, 3, 33, 1, 333};
 var methods = ProfileFunctions.class
 .getDeclaredMethods();
 for (var m : methods) {
 if(m.isAnnotationPresent(Profile.class)) {
 m.setAccessible(true); // best practice or
 sth, idk, i write code in actually good languages
 Profiler.profileMethod(testFunctions, m,
 new Object[] {array});
 }
 }
 }
}

CLASS

TODO:

@Target(ElementType.TYPE)
getDeclaredConstructor().newInstance()
...
ATTRIBUTE

TODO:

@Target(ElementType.FIELD)
...
VALIDATION

@Min(value = 18, message = "Age must be ≥ 18")
@Max(value = 99, message = "Age must be ≤ 99")
private int age;
@NotNull(message = "Name cannot be null")
private String name;

ALGORITHMS

- Brute-force
- Greedy
 - Take best option each step (optimize locally)
 - Fast
- Backtracking
 - Trial and error
 - Best overall option
 - Higher compute costs
- Divide and conquer
 - Binary search
- Dynamic programming
 - Recursion optimization?
 - Tabulation

BACKTRACKING

TODO:
slides 87

BIG O NOTATION

- Worst case scenario
- Atomic operations = constant time
- Runtime measured as sum of primitive operations
Notation
$$\widetilde{f} \left(\underbrace{n}_{\text{Size of problem}} \right) = \text{Number of steps}$$

g = complexity class

TODO:
diagram $f \leq cg$

$f(n)$ is $O(g(n))$ if $c > 0 \wedge n_0 \geq 1$
 $f(n) \leq cg(n)$ for $n \geq n_0$

f in order g:

Let $f, g : \mathbb{N} \rightarrow \mathbb{N}$
 $n_0, c \in \mathbb{N} \wedge \forall n \geq n_0 : f(n) \leq c \cdot g(n)$
 $\Rightarrow f \in O(g) \Leftrightarrow f = O(g)$

Big O

$$\overset{\text{Big O}}{\widetilde{O}} \left(\underbrace{n}_{\text{Size of problem}} \right) = \text{Complexity class}$$

RULES

If $f(n)$ is polynomial of degree d , then $f(n) \in O(n^d)$
- ignore lower powers
- ignore constants
Use most optimal function (lowest power)
- $2n$ is $O(n)$ better than $2n$ is $O(n^2)$
Simplify as far as possible
- $3n + 5$ is $O(n)$ instead of $3n + 5$ is $O(3n)$

COMPLEXITY CLASSES

Name	Class	Example
Constant	1	Array read
Logarithmic	$\log n$	Binary search
Linear	n	Linear search
Log-linear	$n \log n$	Merge+Quick sort
Quadratic	n^2	Bubble+Insertion sort
Cubic	n^3	Solve equation
Polynomial	n^k	Various algorithms
Exponential	2^n	Calculate all subsets
Factorial	$n!$	Calculate all permutations

$O(1)$
 $O(\log n)$
 $O(n)$
 $O(n \log n)$
 $O(n^2)$
 $O(n^3)$
 $O(2^n)$
 $O(2^n)$

TODO:
Counting primitive operations (slides 55)

SORTING ALGORITHMS

SELECTION SORT

- Search for smallest elem in unsorted part and move in front

- Georgiy Shevoroshkin
- Less mutations in array
 - Same performance even if list almost sorted

```
public static void selectionSort(int[] a) {
    int n = a.length;
    for (int i = 0; i < n - 1; i++) {
        int minimum = i;
        for (int j = i + 1; j < n; j++)
            if (a[j] < a[minimum])
                minimum = j;
        swap(a, i, minimum);
    }
}
```

$$(n-1) + (n-2) + \dots + 1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2-n}{2} \sim O(n^2)$$

- INSERTION SORT**
- Take elem and put into correct place in sorted part
 - Good for already partially sorted arrays

```
public static void insertionSort(int[] a) {
    int n = a.length;
    for (int i = 1; i < n; i++)
        for (int j = i; j > 0 && a[j] < a[j - 1]; j--)
            swap(a, j, j - 1);
}
```

$$1 + \dots + (n-1) + n = \sum_{i=1}^n i = \frac{n(n+1)}{2} = \frac{n^2+n}{2} \sim O(n^2)$$

- BUBBLE SORT**
- Compare neighboring elems and swap if needed

```
public static void bubbleSort(int[] a) {
    for (n = a.length; n > 1; n--)
        for (i = 0; i < n - 1; i++)
            if (a[i] > a[i + 1])
                swap(a, i, i + 1);
}
```

$$(n-1) + (n-2) + \dots + 1 = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2} = \frac{n^2-n}{2} \sim O(n^2)$$

```
public static void countingSort(int[] a) {
    int k = Arrays.stream(arr).max().getAsInt();
    int[] c = new int[k + 1];
    for (int i : a) c[i] += 1;
    int pos = 0;
    for (int i = 0; i < k; i++)
        for (int j = 0; j < c[i]; j++) {
            a[pos] = i;
            pos++;
        }
}
```

$$n + n + k + n = 3n + k \sim O(n + k)$$

- SHELL SORT**
- Sort pairs of far away elems
 - Sort the partially sorted array through insertion sort

```
public static void shellSort(int[] a) {
    int n = a.length;
    int h = 1;
    while (h < n/3) h = 3 * h + 1;
    while (h >= 1) {
        for (int i = h; i < n; i++)
            for (int j = i; j >= h && a[j] < a[j - h]; j = j - h) swap(a, j, j - h);
        h /= 3;
    }
}
```

BINARY SEARCH

```
public static <T extends Comparable<T>> boolean bs(
    List<T> data, T target, int low, int high) {
    if (low > high) return false;
    int pivot = low + ((high - low) / 2);
    if (target.equals(data.get(pivot)))
        return true;
    else if (target.compareTo(data.get(pivot)) < 0)
        return searchBinary(data, target, low, pivot - 1);
    else
```

```
return searchBinary(data, target, pivot + 1, high);
}
```

$$O(\log n)$$

- RECURSION**
- When a function calls itself
- LINEAR RECURSION**
- Recursive call starts **at most one** further recursive call

TODO:
recursive → iterative (slides 61)

- TAIL RECURSION**
- Linear recursion where the recursive call is **last**

TODO:
recursive → iterative (slides 69 – nice)

- BINARY RECURSION**
- All non-terminal calls have **two** recursive calls

TODO:
example (slides 76,77,78)

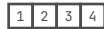
TODO:
 $O(n)$ (slides 84)

DATA STRUCTURES

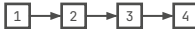
ADT: Abstract Data Type (e.g. Interface)

TODO:
diagram

ARRAY



LINKED LIST



DOUBLY LINKED LIST



STACK

LIFO

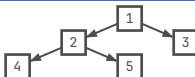


QUEUE

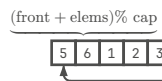
FIFO



TREE



RING BUFFER



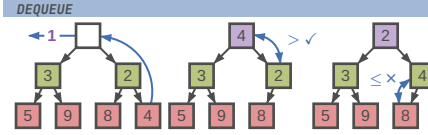
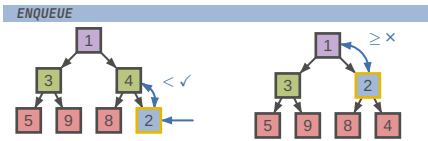
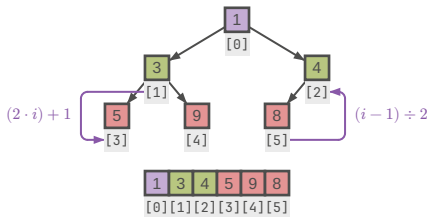
PRIORITY QUEUE

Method	Unsorted List	Sorted List
insert	$O(1)$	$O(n)$
min	$O(n)$	$O(1)$
removeMin	$O(n)$	$O(1)$

HEAP

- Binary tree
- Layers 0, ..., $h-1$ fully populated, layer h filled from left
- Operations always lowest RHS so that tree stays consistent

Min-Heap: Keys of nodes are **smaller** or equal than children's
Max-Heap: Keys of nodes are **larger** or equal than children's



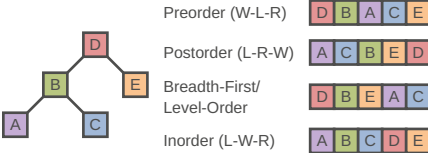
HEAP SORT
Dequeue elems from head into list, resorting tree on each iter

TODO:

BINARY SEARCH TREE (BST)

- All subnodes of **left** subtree are **smaller** than root
- All subnodes of **right** subtree are **larger** than root

TRAVERSING



TODO:

MORE DATA STRUCTURES

- Set:** Elements of same type, no duplicates, without order
- Multiset:** Set with duplicates
- Map:** KV-pairs with unique keys
- Multimap:** Key can have multiple values

HASHING

Hash-function h maps e onto $[0, n-1]$. $h(e)$ = position of e

$$e \xrightarrow{\text{hash code}} \mathbb{Z} \xrightarrow{\text{compression function}} \overbrace{\text{Index}}^{[0, n-1]}$$

Properties of good hash functions:

- Equal distribution
- Fast computation
- Constant time complexity
- Deterministic
- Incorporates as much information from key as possible

HASH FUNCTIONS

- Modulo primes
- Memory address (Object.hashCode() default)
- Byte/Integer cast ByteBuffer.wrap(b).getInt()
- Component sum (eg. sum string char codepoints)
- Polynomial accumulation $a_0 + a_1z + a_2z^2 + \dots + a_{n-1}z^{n-1}$
- $z = 33$ for strings
- Horner-Schema (identical, but easier to compute):
 $- a_0 + z \cdot (a_1 + z \cdot (a_2 + \dots + z \cdot a_{n-1}))$

EQUALITY

It is generally necessary to override hashCode whenever equals is overridden

```
x.equals(y) => x.hashCode() == y.hashCode()
```

HASHING OBJECTS

```
// Horner schema with z = 31 and start = 17
@Override
public int hashCode() {
    int result = 17;
    for (Object field : fields) result = 31 * result
        + (field == null ? 0 : field.hashCode());
    return result;
}
```

ANOMALIES

- Empty spaces
- Collisions
- Overflow

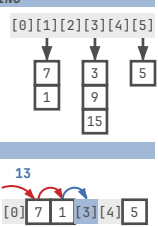
MANAGING HASH COLLISIONS

CLOSED ADDRESSING/SEPARATE CHAINING

Each bucket is a list.

Search: $O(\text{entries}/\text{buckets})$

- Larger storage overhead
- Acceptable performance under high load



OPEN ADDRESSING

e "overflows" into next empty bucket
→ Must probe **2** times in example



TODO:

Problem: Primary Clustering

Linear probing: $s(k, i) = h(k) + i$
Linear probing backwards: $s(k, i) = h(k) - i$
Quadratic probing: $s(k, i) = h(k) + i^2$

TODO:

diagram, slides 10,11

TODO:

deletion (slides 16-21)

- Only mark as "deleted"
- Move next best candidate to deleted position

TODO:

access (slides 22-27)

Probability for finding free spot:

$$n = \text{entries}, N = \text{buckets}$$
$$a = \frac{n}{N} = \text{probability: occupied}, p = \text{probing steps}$$
$$P(X = p) = a^{p-1}(1-a)$$
$$E(X) = \sum_{x=1}^{\infty} p \cdot P(X = p) = \frac{1}{1-a} = \text{Nr. accesses}$$

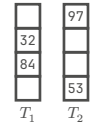
- Resize/Rehash compute intensive
- No storage overhead
- Bad performance under high load

CUCKOO-HASHING

$$h_1(x) = x \bmod 5, h_2(x) = x/5 \bmod 5$$

Insert: If $h_1(x)$ = null $\Rightarrow T_1$, else push y into T_2
If no cycle $\rightarrow O(1)$, if rehash is needed $\rightarrow O(n)$

Search: $O(1) \Rightarrow$ only has to check 2 positions
Rehashing: $x \bmod n \Rightarrow x \bmod (n+m)$



TODO:

EXTENSIBLE HASHING

TODO:

- Resize/Rehash easier
- Larger storage overhead
- Good performance under high load

```
// getting index
int idx = key.hashCode() & ((1 << globalDepth) - 1);
```

```
// when to rehash
if (bucket.getLocalDepth() == globalDepth)
    growHashTable();
else
    splitBlock();
// splitting block
int highBit = 1 << localDepth;
for (Entry e : page.entries) {
    int h = e.key.hashCode();
    Page newPage = (h & highBit) == 0 ? p1 : p0;
    newPage.put(e.key, e.value);
}
```

TODO:
slides 64

